05-19 第1篇

阅读理解（说明文） How Do Chameleons Change Their Colour?

① Among the most famous "quick-change artists" in nature, few can match the **chameleon**. This small lizard lives mainly in warm forests in Africa, Madagascar and parts of southern Asia. Many children grow up believing that a chameleon simply "copies" the colour of any surface it stands on — like a magical living paintbrush. The real story, however, is more interesting and a little different.

② First, chameleons do not change colour only to hide. In fact, hiding is not even their main reason for changing. Scientists who study chameleons have found that they use colour mostly to show their **feelings and messages**. A bright, warm colour may mean "I am angry, stay away." A pale, soft colour may mean "I am calm and willing to be friends." Before a thunderstorm, a chameleon may turn darker so that it can soak up more heat from the last rays of sun.

③ So how do they do it? For many years, people thought it was because of coloured dye in their skin. The truth is quite different. Just under the skin, a chameleon has special cells filled with **tiny crystals**. When the chameleon is calm, these crystals lie close together. Light bouncing off the crystals shows cool colours, such as green and blue. When the chameleon is excited, tiny muscles stretch the skin and pull the crystals slightly apart. Now the same light shows warm colours — yellow, orange, red.

④ This kind of colour is called **structural colour**. A butterfly wing, a peacock feather, and even a thin layer of oil on water all show colour in this way. The colour you see does not come from a paint, but from the way tiny shapes bend and reflect light. Chameleons are special because they can **control** their structures on purpose, many times a day.

⑤ Different kinds of chameleons also have different "colour vocabularies". Panther chameleons, for example, can shine in ten or more colours, while some smaller species change only between a few shades of brown and green. Males usually change more brightly than females, because colour is also used to attract mates. Young chameleons are more limited; their colour skills grow clearer as they grow older.

⑥ Finally, colour change does not happen all at once. It takes about **twenty seconds** for a chameleon to make a full change. So the next time you see a short cartoon where a chameleon turns red the moment it steps on a red hat, remember: real chameleons are artists, but they are gentle artists. Their changing colours are a language, a mood diary written in small shining crystals just under the skin.

根据短文内容，选择最佳答案。

1. What do many children think is the reason chameleons change colour?

A. To sleep better.
B. To simply copy the colour of any surface they stand on.
C. To confuse other animals with sounds.
D. To help humans with paint.

2. What is the MAIN reason chameleons change colour, according to scientists?

A. To hide from predators.
B. To show feelings and messages.
C. To change their weight.
D. To turn into different species.

3. What is hidden just under a chameleon's skin?

A. Small dye bottles.
B. Special cells filled with tiny crystals.
C. Small batteries.
D. Hot air.

4. What is "structural colour"?

A. Colour made by the shape and arrangement of tiny structures reflecting light, not by dye.
B. Colour painted by the chameleon's owner.
C. Colour made by eating spicy food.
D. Colour that only appears in the rain.

5. Which is TRUE according to the passage?

A. A chameleon can turn red in a single second.
B. Chameleons can change colour with the help of tiny crystals and muscles in about 20 seconds.
C. Young chameleons have the best colour skills.
D. All chameleons have the same colour vocabulary.

答案与解析

1.
【答案】B
【题型】细节理解题

【定位】第①段末 "a chameleon simply 'copies' the colour of any surface..."
【解析】孩子们以为变色龙只是复制颜色。A/C/D 均不符。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第②段第2–3句 "they use colour mostly to show their feelings and messages."
【解析】主要是表达情绪与信息。A 是次要，C/D 均错误。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第③段 "Just under the skin, a chameleon has special cells filled with tiny crystals."
【解析】皮下有含微小晶体的细胞。A/C/D 均不符。故选 B。

4.
【答案】A
【题型】词义猜测题
【定位】第④段 "from the way tiny shapes bend and reflect light"
【解析】结构色由微小结构反射/折射光形成，不靠颜料。B/C/D 均不符。故选 A。

5.
【答案】B
【题型】推理判断题
【定位】第③段晶体+肌肉机制；第⑥段"about twenty seconds"
【解析】变色需约 20 秒，借晶体与肌肉完成。A 与事实相反，C/D 与原文矛盾。故选 B。

词汇注释（超纲词）

- **chameleon** [n.] 变色龙
- **crystal** [n.] 晶体
- **structural colour** [n. phr.] 结构色
- **species** [n.] 物种